Weekly Record W4: Human-AI Operating Model & Task Allocation Framework

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Company A: Shell

Company B: KazMunayGas (KMG)

1. Academic & Theoretical Foundation

The integration of Artificial Intelligence into corporate operations requires a structural redefinition of organizational workflows. Grounded in organizational design theory, work design must balance algorithmic automation with human cognitive oversight. The Human-in-the-Loop (HITL) architecture categorizes operational tasks along a continuum of autonomy based on risk profile, decision complexity, and legal liability.

For high-hazard processing environments such as petroleum refineries and offshore platforms, delegating complete decision-making authority to autonomous algorithms introduces catastrophic operational and legal risks. Consequently, an effective Human-AI Operating Model establishes explicit boundaries where machines execute data processing and anomaly detection, while named human managers retain absolute legal accountability for operational interventions, safety shutdowns, and strategic capital distribution.

2. Practical Artifact: 8 - Task Human-AI Allocation Matrix for KMG

Table 4.1: Human-AI Task Allocation Framework

Task #	Operational Task Description	Risk Profile	Human-AI Allocation Category	Operational Mechanism & Governance Protocol
Task 1	Refinery Sensor Data Ingestion	Low	**Automated**	Continuous IoT telemetry logging without human intervention
Task 2	Anomaly Detection in Pumps	Low-Med	**AI Executes + Review**	Algorithm flags vibration deviations; technician reviews alert
Task 3	Water-Flooding Rate Adjustment	Medium	**AI Executes + Review**	ABAI calculates injection rates; reservoir engineer approves

Task 4	Emergency Refinery Unit Shutdown	High	**Human Only**	Plant Director holds sole legal authority to initiate shutdown
Task 5	Wholesale Fuel Market Analysis	Medium	**Human + AI**	AI models supply-demand trends; commercial team sets strategy
Task 6	Capital Expenditure Allocation	High	**Human Only**	Management Board and Board of Directors retain approval power
Task 7	Regulatory Compliance Logging	Low	**Automated**	Automated generation of environmental compliance reports
Task 8	Regional Labor & HR Policy	High	**Human Only**	Executives and union reps negotiate employment terms

3. Comparative Analysis: Shell vs. KazMunayGas (KMG)

1. **Automation Autonomy**: Shell deploys highly autonomous closed-loop control systems in modern assets (e.g., Pernis refinery, deepwater Gulf of Mexico), where AI algorithms directly adjust processing parameters under exception-based human monitoring. KMG enforces a stricter HITL regime, requiring human engineering confirmation prior to process parameter changes.
2. **Accountability & Governance**: Shell distributes operational accountability across global matrix management structures. KMG maintains centralized, legally designated operational roles (e.g., Chief Engineer, Refinery Technical Director) who bear personal civil and criminal liability for plant safety under Kazakh industrial laws.
3. **Workforce Impact**: Shell utilizes algorithmic task allocation to reduce onshore operational headcount. KMG uses AI to augment human capabilities, prioritizing safety enhancement and equipment longevity over headcount reduction.

4. Strategic Transferability Verdict

- **ADAPT**: Adapt Shell's structured task-taxonomy methodology to systematically audit and categorize operational workflows across all KMG subsidiaries.
- **REJECT**: Reject autonomous, closed-loop algorithmic execution for high-hazard refinery interventions; enforce mandatory human approval protocols for all critical operational control loops.

5. AI Verification & Level 2 Declaration Log

- **Prompt Used (Gemini 2.5 Pro)**: *"Categorize operational tasks for an oil refinery into an automated task matrix, including emergency shutdown procedures."*
- **AI Hallucination Identified**: The AI model classified "Emergency Refinery Unit Shutdown" under **Fully Automated**, arguing that algorithms react faster than human operators to pressure spikes.
- **Human Verification & Correction**: Reclassified to **Human Only**. In a high-hazard environment like Atyrau Refinery (ANPZ), an unexpected automated shutdown can trigger downstream systemic shocks, emergency flaring, or supply deficits. A named human engineer must evaluate context before authorizing a shutdown.

6. Actionable Operational Recommendations & Key Metrics

1. **Governance Rule**: Formally embed the 8-Task Human-AI Matrix into KMG's Operational Safety Standard Operating Procedures (SOPs).
2. **Training Metric**: Certify 100% of refinery plant operators and control room personnel on HITL override procedures by Q2 2027.
3. **Audit Protocol**: Conduct semi-annual reviews of AI recommendation acceptance rates among plant engineers to detect model distrust or over-reliance.

7. Reflection, Peer Review & Capstone Linkage

This task allocation framework served as the core analytical deliverable for Attestation A2 (Video Essay & Evidence File) and directly provides the operational governance structure displayed in Capstone (Human-AI Operating Model).